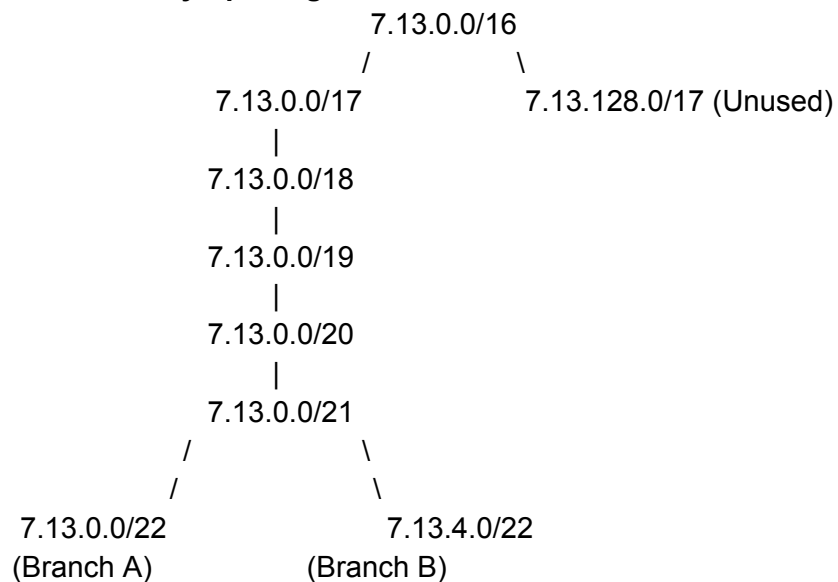


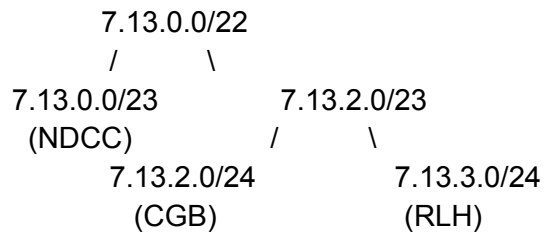
VLSM Calculation Table

Segment / Unit	Host Req.	Needed Address (+2)	Block Size (2^n)	Host Bits	Prefix
NDCC	280	282	512	9	/23
CGB	200	202	256	8	/24
RLH	220	222	256	8	/24
BMD	150	152	256	8	/24
FRC	130	132	256	8	/24
TRP	90	92	128	7	/25
COI	40	42	64	6	/26
Central Switch	4	6	8	3	/29
NDCC <-> RLH	2	4	4	2	/30
NDCC <-> BMD	2	4	4	2	/30
BMD <-> CGB	2	4	4	2	/30
CGB <-> NDCC	2	4	4	2	/30
RLH <-> FRC	2	4	4	2	/30
TRP <-> COI	2	4	4	2	/30

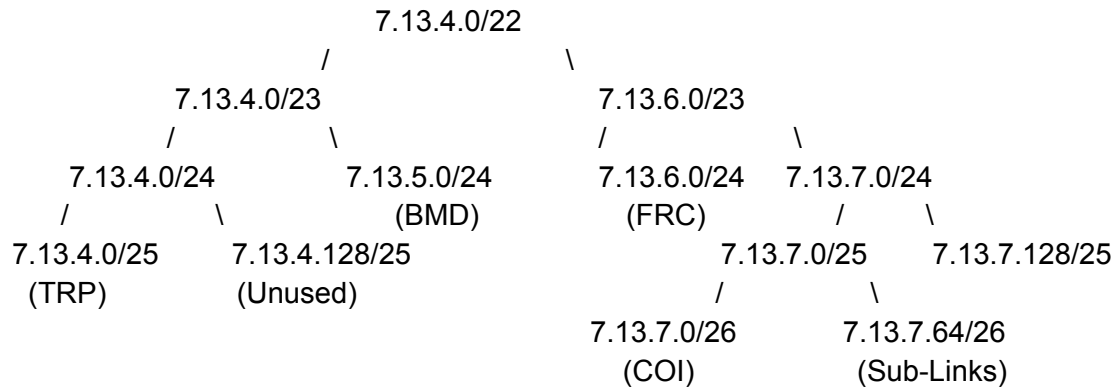
Hierarchical VLSM Binary Splitting Tree:



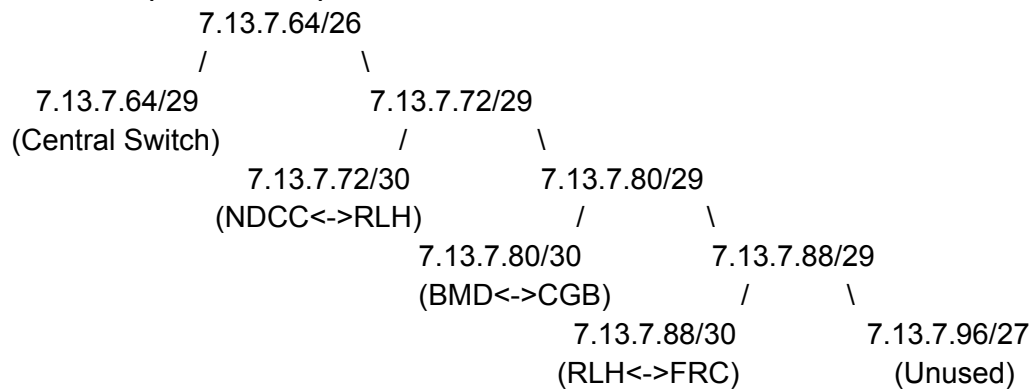
Branch A Division (7.13.0.0/22)



Branch B Division (7.13.4.0/22)



Sub-Links Division (7.13.7.64/26)



Subnet Allocation Table

	Segment	Network / Prefix	Subnet Mask	Usable IP Range	Broadcast
1	NDCC	7.13.0.0/23	255.255.254.0	7.13.0.1 - 7.13.1.254	7.13.1.255
2	CGB	7.13.2.0/24	255.255.255.0	7.13.2.1 - 7.13.2.254	7.13.2.255
3	RLH	7.13.3.0/24	255.255.255.0	7.13.3.1 - 7.13.3.254	7.13.3.255
4	TRP	7.13.4.0/25	255.255.255.128	7.13.4.1 - 7.13.4.126	7.13.4.127
5	BMD	7.13.5.0/24	255.255.255.0	7.13.5.1 - 7.13.5.254	7.13.5.255
6	FRC	7.13.6.0/24	255.255.255.0	7.13.6.1 - 7.13.6.254	7.13.6.255
7	COI	7.13.7.0/26	255.255.255.192	7.13.7.1 - 7.13.7.62	7.13.7.63
8	Central Switch	7.13.7.64/29	255.255.255.248	7.13.7.65 - 7.13.7.70	7.13.7.71
9	NDCC <-> RLH	7.13.7.72/30	255.255.255.252	7.13.7.73 - 7.13.7.74	7.13.7.75
10	NDCC <-> BMD	7.13.7.76/30	255.255.255.252	7.13.7.77 - 7.13.7.78	7.13.7.79
11	BMD <-> CGB	7.13.7.80/30	255.255.255.252	7.13.7.81 - 7.13.7.82	7.13.7.83
12	CGB <-> NDCC	7.13.7.84/30	255.255.255.252	7.13.7.85 - 7.13.7.86	7.13.7.87
13	RLH <-> FRC	7.13.7.88/30	255.255.255.252	7.13.7.89 - 7.13.7.90	7.13.7.91
14	TRP <-> COI	7.13.7.92/30	255.255.255.252	7.13.7.93 - 7.13.7.94	7.13.7.95

NDCC

Basic Setup

```
enable
configure terminal
hostname NDCC
```

Interface Addressing

```
interface GigabitEthernet0/0
ip address 7.13.0.1 255.255.254.0
no shutdown
exit
```

```
interface GigabitEthernet0/1
ip address 7.13.7.65 255.255.255.248
no shutdown
exit
```

```
interface Serial0/1/0
ip address 7.13.7.73 255.255.255.252
clock rate 64000
no shutdown
exit
```

```
interface Serial0/1/1
ip address 7.13.7.77 255.255.255.252
clock rate 64000
no shutdown
exit
```

```
interface Serial0/2/0
ip address 7.13.7.86 255.255.255.252
no shutdown
exit
```

DHCP Configuration (Member 1) — server for NDCC, BMD, CGB (Warning Loop)

```
ip dhcp excluded-address 7.13.0.1 7.13.0.9
ip dhcp excluded-address 7.13.5.1 7.13.5.9
ip dhcp excluded-address 7.13.2.1 7.13.2.9
```

```
ip dhcp pool NDCC-POOL
network 7.13.0.0 255.255.254.0
```

```
default-router 7.13.0.1
dns-server 7.13.0.5
exit
```

```
ip dhcp pool BMD-POOL
network 7.13.5.0 255.255.255.0
default-router 7.13.5.1
dns-server 7.13.0.5
exit
```

```
ip dhcp pool CGB-POOL
network 7.13.2.0 255.255.255.0
default-router 7.13.2.1
dns-server 7.13.0.5
exit
```

Static Routing (Member 2) — routes to RLH/TRP/FRC/COI + backup to RLH

```
ip route 7.13.3.0 255.255.255.0 Serial0/1/0
ip route 7.13.4.0 255.255.255.128 7.13.7.68
ip route 7.13.6.0 255.255.255.0 7.13.7.74
ip route 7.13.7.0 255.255.255.192 7.13.7.68
ip route 7.13.3.0 255.255.255.0 7.13.7.67 5
```

Dynamic Routing (Member 3) — RIPv2 + redistribution

```
router rip
version 2
redistribute static metric 1
network 7.0.0.0
no auto-summary
exit
```

Save Configuration

```
end
copy running-config startup-config
```

CGB

Basic Setup

```
enable
configure terminal
hostname CGB
```

Interface Addressing + DHCP Relay (Member 1) — relays to NDCC

```
interface GigabitEthernet0/0
ip address 7.13.2.1 255.255.255.0
ip helper-address 7.13.0.1
no shutdown
exit
```

```
interface GigabitEthernet0/1
ip address 7.13.7.66 255.255.255.248
no shutdown
exit
```

```
interface Serial0/1/0
ip address 7.13.7.82 255.255.255.252
no shutdown
exit
```

```
interface Serial0/1/1
ip address 7.13.7.85 255.255.255.252
no shutdown
exit
```

Dynamic Routing (Member 3) — RIPv2

```
router rip
version 2
network 7.0.0.0
no auto-summary
exit
```

Save Configuration

```
end
copy running-config startup-config
```

RLH

Basic Setup

```
enable
configure terminal
hostname RLH
```

Interface Addressing

```
interface GigabitEthernet0/0
ip address 7.13.3.1 255.255.255.0
no shutdown
exit
```

```
interface GigabitEthernet0/1
ip address 7.13.7.67 255.255.255.248
no shutdown
exit
```

```
interface Serial0/1/0
ip address 7.13.7.74 255.255.255.252
no shutdown
exit
```

```
interface Serial0/1/1
ip address 7.13.7.89 255.255.255.252
clock rate 64000
no shutdown
exit
```

DHCP Configuration (Member 1) — own server, also serves FRC via relay

```
ip dhcp excluded-address 7.13.3.1 7.13.3.9
ip dhcp excluded-address 7.13.6.1 7.13.6.9
```

```
ip dhcp pool RLH-POOL
network 7.13.3.0 255.255.255.0
default-router 7.13.3.1
dns-server 7.13.0.5
exit
```

```
ip dhcp pool FRC-POOL
network 7.13.6.0 255.255.255.0
default-router 7.13.6.1
```

```
dns-server 7.13.0.5
exit
```

Static Routing (Member 2) — routes to every external network + recursive backup to NDCC

```
ip route 7.13.6.0 255.255.255.0 Serial0/1/1
ip route 7.13.0.0 255.255.254.0 Serial0/1/0
ip route 7.13.2.0 255.255.255.0 7.13.7.66
ip route 7.13.4.0 255.255.255.128 7.13.7.68
ip route 7.13.5.0 255.255.255.0 7.13.7.65
ip route 7.13.7.0 255.255.255.192 7.13.7.68
ip route 7.13.0.0 255.255.254.0 7.13.7.68 5
```

Save Configuration

```
end
copy running-config startup-config
```

TRP

Basic Setup

```
enable
configure terminal
hostname TRP
```

Interface Addressing

```
interface GigabitEthernet0/0
ip address 7.13.4.1 255.255.255.128
no shutdown
exit
```

```
interface GigabitEthernet0/1
ip address 7.13.7.68 255.255.255.248
no shutdown
exit
```

```
interface Serial0/1/0
ip address 7.13.7.93 255.255.255.252
clock rate 64000
no shutdown
exit
```


DHCP Configuration (Member 1) — own server, also serves COI via relay

```
ip dhcp excluded-address 7.13.4.1 7.13.4.9
ip dhcp excluded-address 7.13.7.1 7.13.7.9
```

```
ip dhcp pool TRP-POOL
network 7.13.4.0 255.255.255.128
default-router 7.13.4.1
dns-server 7.13.0.5
exit
```

```
ip dhcp pool COI-POOL
network 7.13.7.0 255.255.255.192
default-router 7.13.7.1
dns-server 7.13.0.5
exit
```

Static Routing (Member 2) — routes to every external network + floating backup to NDCC

```
ip route 7.13.7.0 255.255.255.192 Serial0/1/0
ip route 7.13.2.0 255.255.255.0 7.13.7.66
ip route 7.13.3.0 255.255.255.0 7.13.7.67
ip route 7.13.5.0 255.255.255.0 7.13.7.65
ip route 7.13.6.0 255.255.255.0 7.13.7.67
ip route 7.13.0.0 255.255.254.0 7.13.7.66 57
ip route 7.13.0.0 255.255.254.0 7.13.7.65
```

► Save Configuration

```
end
copy running-config startup-config
```

BMD

Basic Setup

```
enable
configure terminal
hostname BMD
```

Interface Addressing + DHCP Relay (Member 1) — relays to NDCC

```
interface GigabitEthernet0/0
ip address 7.13.5.1 255.255.255.0
ip helper-address 7.13.0.1
no shutdown
exit
```

```
interface Serial0/1/0
ip address 7.13.7.78 255.255.255.252
no shutdown
exit
```

```
interface Serial0/1/1
ip address 7.13.7.81 255.255.255.252
clock rate 64000
no shutdown
exit
```

Dynamic Routing (Member 3) — RIPv2 (learns everything via redistribution, zero static routes needed)

```
router rip
version 2
network 7.0.0.0
no auto-summary
exit
```

Save Configuration

```
end
copy running-config startup-config
```

FRC

Basic Setup

```
enable
configure terminal
hostname FRC
```

Interface Addressing + DHCP Relay (Member 1) — relays to RLH

```
interface GigabitEthernet0/0
ip address 7.13.6.1 255.255.255.0
ip helper-address 7.13.3.1
no shutdown
exit
```

```
interface Serial0/1/0
ip address 7.13.7.90 255.255.255.252
no shutdown
exit
```

Static Routing (Member 2) — exit-interface routes to every external network (PDF-required style)

```
ip route 7.13.3.0 255.255.255.0 Serial0/1/0
ip route 7.13.0.0 255.255.254.0 Serial0/1/0
ip route 7.13.2.0 255.255.255.0 Serial0/1/0
ip route 7.13.4.0 255.255.255.128 Serial0/1/0
ip route 7.13.5.0 255.255.255.0 Serial0/1/0
ip route 7.13.7.0 255.255.255.192 Serial0/1/0
```

Save Configuration

```
end
copy running-config startup-config
```

COI

Basic Setup

```
enable
configure terminal
hostname COI
```

Interface Addressing + DHCP Relay (Member 1) — relays to TRP

```
interface GigabitEthernet0/0
ip address 7.13.7.1 255.255.255.192
ip helper-address 7.13.4.1
no shutdown
exit
```

```
interface Serial0/1/0
ip address 7.13.7.94 255.255.255.252
no shutdown
exit
```

Static Routing (Member 2) — default route (COI's only path out is via TRP)

```
ip route 0.0.0.0 0.0.0.0 Serial0/1/0
```

Save Configuration

```
end
copy running-config startup-config
```